

ARIF MEHMOOD

AI ENGINEER · LLM APPLICATIONS · BACKEND & CLOUD · PRODUCTION SYSTEMS SINCE 2018

Redditch, UK · Full UK right to work · UK Global Talent visa · no sponsorship required

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PROFILE

In software engineering since 2018. I have shipped a mobile payments app that grew to around 20,000 users, point-of-sale software running in high-volume retail chains, and a data platform live to 10 UK universities that I released fortnightly for two years. I now build the AI and backend of a consumer health app, and I am the technical point of contact for product, QA, DevOps and the commercial teams.

CORE SKILLS

AI & LLM Engineering: LLM application design, prompt engineering, structured output validation, grounded generation with deterministic retrieval, fallback and graceful degradation, prompt-injection mitigation, PII minimisation in prompts, self-hosted inference (Ollama, vLLM), hosted model APIs (Anthropic), LLM logging and cost instrumentation, scikit-learn, SHAP explainability

Backend, Cloud & Data: Python, FastAPI REST services, Java 17, Spring Boot, Spring Cloud microservices, REST API design, JWT authentication, PostgreSQL schema design, DynamoDB, OpenSearch, AWS (Lambda, API Gateway, Cognito, Amplify, EC2), Azure container hosting, Docker containerisation, GitHub Actions CI/CD workflows, SonarQube quality gates, Linux.

Cross-functional & Delivery Ownership: Release test planning and QA sign-off criteria, infrastructure requirements and cost options with DevOps, external supplier evaluation and procurement recommendations, competitor and market review, production support and incident ownership, mentoring and code review, introducing Agile process and Jira

EXPERIENCE

AI Engineer — BioCare Ltd (in partnership with Aston University)

Jan 2026 – Present

Redditch, UK

Product: A consumer health app: someone answers a health questionnaire and gets back a personalised report scoring nine areas of their health, with nutrition and lifestyle guidance written for their own answers. **In user acceptance testing; launch targeted January 2027.**

My scope: The engineer responsible for the AI and backend of the product, and the technical point of contact across product, QA, DevOps, brand, marketing and sales.

Technologies: Python · FastAPI REST services · PostgreSQL schema design · Docker containerisation · Anthropic API (hosted LLM inference) · Ollama and vLLM (self-hosted LLM inference) · scikit-learn classification · SHAP model explainability · React · GitHub Actions CI/CD workflows · Azure container hosting

- **Build the part of the app that explains a person's results.** Rules choose the guidance from a library written by the company's nutrition team; a language model then rewords it for the individual, so it can change how something is said but never what is recommended. If a model call fails, that section falls back to the expert wording and the report still completes.
- **Build the scoring engine at the centre of the product.** It turns a 300-question assessment into ranked scores across nine areas of health, using about 1,250 weighted question-to-area mappings from the company's expert model. The same answers always give the same result, and automated tests check the code against the nutritionists' reference spreadsheet.
- **Set the privacy rules for anything that touches the model.** No name or email address goes into a prompt, anything a user typed is handled as data rather than as instructions, and every outgoing request is logged for audit with credentials stripped.
- **Redirected the product, then delivered the replacement.** The brief was machine-learning-first. I prepared around 56,000 public health survey records and built nine classifiers with explainability, and the prototype I demonstrated in March 2026 showed the data came from a US population with gaps that would not give UK users reliable results. I made the case for an expert-rules product that keeps machine learning open for later and reframed the outputs so the app scores wellbeing rather than implying diagnoses.
- **Act as the technical point of contact across the business.** I own the release test plan and agree with QA what counts as working, set out what the platform needs from DevOps and work through the cost of the options with them before we decide, run the external supplier evaluations that go to leadership, and translate the app's scores into terms brand, marketing and sales can sell against.

Software Engineer — Cardiff University

Dec 2023 – Jan 2026

Cardiff, UK

Product: A shared data and reporting platform that 10 universities across Wales use to submit, transform and report on shared datasets, plus three further systems for government and education customers. **Live in production; released fortnightly for two years.**

My scope: Architecture and delivery owner on the university platform, developer on three further systems, and the person who trained the team that took it over.

Technologies: Python · FastAPI · Django · Java 17 · Spring Boot APIs · React · Node.js · AWS Lambda event-driven workflows · AWS Cognito authentication · OpenSearch indexing and search · DynamoDB · Kibana reporting dashboards · Docker packaging · GitLab CI release automation · Jira

- **Owned the university data platform end to end.** Designed it, built it on AWS with event-driven processing, role-based access and reporting dashboards, released it to 10 institutions and handed it over. My decisions, my releases, my problems to fix.
- **Released fortnightly for two years without taking the service down.** Ran two versions side by side and signalled connected clients to reload after each release, so fixes shipped in working hours rather than in a maintenance window.

- **Introduced a way of delivering to a project that had none.** Brought in Jira and a working sprint and release process, documented the architecture, and trained colleagues to run, demonstrate and support the platform so it stopped depending on the person who wrote it.
- **Delivered three more systems and taught the department's Generative AI course.** A transport analytics platform for Welsh Government transport stakeholders, a data-sharing platform used by university linguistics teams that featured in a NAACL 2025 demonstration, and extensions to a cybersecurity training platform.

Senior Software Engineer — Confiz Pvt Ltd

Mar 2021 – Sep 2022

Product: A mobile payments app for consumers, with a back-office portal the support team used to run it. **Live in production; grew from 2,000 beta users to around 20,000.**

My scope: Senior backend engineer on the payments platform. Led the production support team, led back-office delivery, and mentored two junior engineers.

Technologies: Java 17 · Spring Boot · Spring Cloud microservices · service discovery and API gateway · React · PostgreSQL transactional design · ELK centralised logging · AWS EC2 · Docker image builds · SonarQube quality gates · GitHub Actions build pipelines

- **Built the backend services and APIs behind a payment's app in production.** Improved system performance by 15% while the user base grew tenfold.
- **Split the monolith into seven services,** separating telecom-operator integrations, identity verification and login, so each integration could be released on its own.
- **Led the production support team,** chosen for it because I knew the platform best. Pulled development, QA and the customer support desk into one diagnosis loop, which took typical resolution from three or four days to inside 24 hours.
- **Rebuilt the support team's back-office portal in React,** to 87% positive feedback from the people who use it daily, then made its adverts and offers configurable so support could run campaigns themselves, cutting the time per change by 98%.
- **Made code quality a gate rather than a report nobody read,** putting SonarQube into the build pipeline: code hotspots to zero and deployment failures down 60%. Mentored two junior engineers through their first production work.

Software Engineer (SDET) — Afiniti

Apr 2020 – Mar 2021

Product: AI software that decides how incoming customer calls are paired with contact-centre agents, sold to large enterprises. **Live in production at enterprise customers.**

My scope: Test and release automation for the AI products, working alongside the QA and data teams.

Technologies: Python test automation · JavaScript · synthetic data generation · Docker test environments · Linux · functional, unit and performance testing · Scrum and Kanban delivery

- **Generated synthetic test data so the AI could be tested on cases the real data did not contain.** Wrote Python to take available volume up tenfold with controlled variation, which is how edge-case behaviour got exercised before release rather than after.
- **Built the automation that tested both the AI products and the release process itself,** raising test coverage by 25% and deployment reliability by 40%.

Software Engineer — Flooid

Jun 2019 – Apr 2020

United Kingdom

Product: Point-of-sale and stock management software for retailers. **Live in production in high-volume retail chains.**

My scope: Developer on till and inventory features, and on the tests and build pipeline that shipped them. Stood in as Scrum Master when the team needed unblocking.

Technologies: Java 8 · Spring Core and Spring Boot · JUnit and Mockito unit testing · Selenium UI test automation · Maven and ANT build scripting · Jenkins CI/CD builds · SonarQube code analysis · Mercurial version control · IBM DB2 · SQL

- **Built and extended till and inventory features** on a Java and Spring platform running in high-volume retail chains, delivering backlog stories inside two-week sprints.
- **Wrote the unit and integration tests and kept the build pipeline green,** clearing SonarQube blockers so defects were caught before acceptance testing rather than after.

Associate Software Engineer — Grid Systems / Global Rescue LLC

Sep 2018 – Jun 2019

Product: Membership and emergency-response systems for a travel risk and medical assistance business. **Live in production.**

My scope: Java developer on reporting, mapping and integration features. Reviewed code and QA test cases.

Technologies: Java EE · EJB · Hibernate ORM · JSF and RichFaces interfaces · MySQL query optimisation · Maven builds · SOAP and REST web services

- **Built and maintained the Java EE, EJB and Hibernate functionality** behind a live membership and emergency-response platform.
- **Turned business and data requirements into the reports the operations team used daily,** writing and optimising the underlying MySQL queries.

EDUCATION & CERTIFICATION

MSc Data Science and Analytics, Distinction — Cardiff University

Sep 2022 – Sep 2023

Machine learning, statistical analysis, neural networks, data wrangling, text processing, databases, visualisation. Python, R.

BEng Computer Engineering — National University of Sciences and Technology (NUST)

Sep 2014 – Jun 2018

AWS Certified Cloud Practitioner

June 2022